

Assessment Task No. 1

Chosen ML Application: Social Media Recommendation

Social Media Platforms have large amounts of content making it difficult for a user to find content based on their interest. Using a social media recommendation system it can solve this by analyzing the user activity.

Part A - Identify the ML Problem

Question	Answer
1. What is the application?	Used by social media platforms to recommend content that the user may be interested in.
2. What problem are you trying to solve?	To match a user's content interest.
3. What are the inputs	User views, Page views, Search history, Likes and follows.
4. What is the expected output?	To recommend content that matches user interests.
5. What patterns or relationships might the ML system learn?	What type of content that the user views or interacts the most.
6. What data would the system need?	User activity, views, and interaction.

Part B - Traditional Programming vs Machine Learning

In traditional programming the programmer would have to manually create the guidelines for what type of content that will be recommended to the user. In social media that means creating a different rule for any possible situation, because of the amount of content that a user consumes on a daily basis it would be difficult and time consuming for the programmer.

But, with machine learning, it can use an algorithm using the user's behavior and the content that the user has interacted with. The algorithm can then identify what type of content that the user will most likely interact with. The machine learning system learns from the patterns that the users make to further recommend content that is suited to their taste.

Part C - Create your own Mini Dataset

User Likes	Recommended Similar Content
1	2
2	4
3	6
4	8
5	10

How could the Machine Learning Algorithm learn this relationship from the examples?

The algorithm would look at what the user likes and recognize what type of content it is and compare it to what other type of content that the user views and it would recommend content similar to what the user had liked. The algorithm can use this to recommend content ($\text{Likes} = \text{Similar Content} \times 2$), it can then make new predictions on what other type of content that the user may be inclined to be interested in.

Part D - Explain the Training Process

The algorithm learns using the examples such as the user likes and similar content, it then compares the amount of likes that the user has on other types of content to recommend it. Then the algorithm checks on what type of content that the user has liked to predict the type of content that the user wants to see. It learns from what the user has liked and makes predictions based on the user. After training the algorithm, it can now predict the content that the user wants to see or what the user may be interested in.

Reflection

The data learned can be useful because real world scenarios can have too many possible combinations. A user may have different interests and can even change over time, writing something to fit every possible user in any situation would require a lot of work. Machine learning allows us and the system to discover patterns instead of manually developing something

to be tailored to a user and as more users in turn have more data, an algorithm can make use of this to make more accurate predictions.